Traffic Monitoring System

https://github.com/aiyshwary/Traffic-Monitoring-System

PHP MySQL JavaScript HTML CSS XAMPP

OVERVIEW

A PHP and SQL-based web application that supports traffic controllers with real-time violator tracking, periodic traffic status notifications, daily reports, and transport department recommendations for infrastructure improvements.

IMPACT

Developed a multi-role Traffic Monitoring System web application with separate portals for admins, officers, and citizens, enabling automated violator tracking, fine management, road signalling status, and daily traffic reporting.

TECHNICAL WALKTHROUGH

Traffic Monitoring System
(TMS)

Date: March 8, 2026
Built with PHP, MySQL, Apache (MAMP) | Bootstrap 4 Frontend

Project Overview
The Traffic Monitoring System (TMS) is a web-based application designed to streamline traffic management for both authorities (Officers) and the public (Citizens). It provides tools for monitoring road conditions, managing traffic signal timings, and handling traffic violation fines - including online payment. The system is built using PHP for server-side logic, MySQL as the relational database, Apache as the web server (via MAMP), and Bootstrap 4 for a responsive frontend UI.

Technology Stack

Layer Technology / Library

1. Backend Language PHP 8.3 (server-side scripting)
2. Database MySQL 8 (via MySQLi extension)
3. Web Server Apache (MAMP on macOS)

Frontend Framework Bootstrap 4

JavaScript Libraries jQuery 2/3, Owl Carousel, ScrollReveal
CSS Libraries Font Awesome 4.7, Animate.css, Perfect Scrollbar
UI Utilities Select2, Tilt.js, Lightbox

Session Management PHP Native Sessions

User Roles

The system has three distinct user roles, each identified by a file prefix convention

Prefix Role Access Level

a Admin / Public Landing page, login/register chooser. No authentication required. c Citizen Registered vehicle owners. Can view own fines, pay fines, view signals and roads. o Officer Traffic authority personnel. Can manage fines, edit signal timings, view road data.

File Structure & Purpose

4.1 Shared / Core Files

File Purpose

aconnection.php MySQL DB connection - included by all PHP pages. Holds host, user, pass, dbname. logout.php Destroys the PHP session and redirects the user back to ahome.php. database.sql Full MySQL schema - creates all 5 tables and inserts sample road/signal data. 4.2 Public / Admin Pages (prefix: a)

File Purpose

ahome.php Public homepage. Slider, About, Features, Team, Contact sections. alioption.php Login chooser page - two buttons: Citizen Login or Officer Login. asuoption.php Register chooser page - two buttons: Citizen Signup or Officer Signup. alogout.php Legacy logout file kept for compatibility.
4.3 Citizen Pages (prefix: c)

File Purpose

csignup.php Registration form. Inserts into 'citizen' table AND auto-creates a fine record. clogin.php Login form. Validates against 'citizen' table, sets $_SESSION['email']. cfunctions.php check_login() function - validates session, redirects to clogin.php if absent. chome.php Citizen dashboard with slider linking to all citizen features. cfine.php Shows own fine records (amount, status). Pay Now button if unpaid. cpayment.php Card payment UI. On submit sets fine.paid = 1 in database. csignalling.php Read-only view of signal timings (road name, time_from, time_to). croad.php Read-only view of all roads (station name, road name).
4.4 Officer Pages (prefix: o)

File Purpose

osignup.php Registration form for officers. Stores username, email, OID, password. ologin.php Login form. Validates against 'officer' table, sets $_SESSION['email']. ofunctions.php check_login() function - validates session, redirects to ologin.php if absent. ohome.php Officer dashboard with slider linking to all officer features. ofine.php Lists ALL fines in the system with an Edit button per row. ozfupdate.php Edit form to update a specific fine amount by fine ID. osignalling.php Lists all signal timings with an Edit button per row. osupdate.php Edit form to update time_from and time_to for a signal entry. oroad.php View all roads - Road ID, Station Name, Road Name.

Database Schema

Database name: 'project' | Character set: utf8mb4 | Engine: InnoDB

Table: citizen - Stores registered citizens

Column Type Constraints

1. id INT AUTO_INCREMENT, PRIMARY KEY
2. username VARCHAR(100) NOT NULL
3. email VARCHAR(100) NOT NULL, UNIQUE
4. Phone VARCHAR(15)

5. VID VARCHAR(20) Vehicle ID

pass VARCHAR(255) NOT NULL (plain text - upgrade to hashed in production)

Table: officer - Stores traffic officers

Column Type Constraints

1. id INT AUTO_INCREMENT, PRIMARY KEY
2. username VARCHAR(100) NOT NULL
3. email VARCHAR(100) NOT NULL, UNIQUE
4. OID VARCHAR(20) Officer ID
5. pass VARCHAR(255) NOT NULL

Table: fine - Traffic violation fines

Column Type Constraints

1. id INT AUTO_INCREMENT, PRIMARY KEY
2. username VARCHAR(100) Links to citizen.username
3. VID VARCHAR(20) Vehicle ID
4. amount DECIMAL(10,2) Default 0.00

paid TINYINT 0 = Pending, 1 = Paid

Table: road - Road information

Column Type Constraints

1. id INT AUTO_INCREMENT, PRIMARY KEY
2. station_name VARCHAR(100)
3. road_name VARCHAR(100)

Table: signalling - Traffic signal timings per road

Column Type Constraints

1. id INT AUTO_INCREMENT, PRIMARY KEY

2. road_name VARCHAR(100)

3. time_from VARCHAR(20) e.g. 06:00

4. time_to VARCHAR(20) e.g. 10:00

Complete Application Flow

6.1 Citizen Flow

Step Page Action

Step 1 Visit ahome.php Public homepage. Click 'Register Now'.
Step 2 asuoption.php Choose 'CITIZEN' to go to csignup.php.
Step 3 csignup.php (POST) Fill name, email, phone, vehicle ID, password. Two DB inserts: citizen table + fine table (amount=0).
Step 4 clogin.php (POST) Enter email + password. Session set on success. Redirect to chome.php.
Step 5 chome.php Dashboard. Links to Fine Payment, Signal Timings, Google Maps, Road Info.
Step 6 cfine.php View own fine. If amount > 0 and unpaid, 'Pay Now' link appears.
Step 7 cpayment.php Enter card details. On submit: fine.paid = 1 in DB. Redirected back with success message.
Step 8 logout.php Session destroyed. Redirect to ahome.php.

6.2 Officer Flow

Step Page Action

Step 1 ahome.php Click 'Register Now'.
Step 2 asuoption.php Choose 'OFFICER' to go to osignup.php.
Step 3 osignup.php (POST) Fill name, email, Officer ID, password. Insert into officer table.
Step 4 ologin.php (POST) Login. Session set. Redirect to ohome.php.
Step 5 ohome.php Dashboard. Links to Fine Allotment, Signal Timings, Google Maps, Road Info.
Step 6 ofine.php View ALL citizen fines. Each row has an Edit link.
Step 7 ozfupdate.php Edit fine amount for a specific citizen. Updates fine.amount in DB.
Step 8 osignalling.php View all signal timings. Each row has an Edit link.
Step 9 osupdate.php Update time_from and time_to for a road signal.
Step 10 oroad.php View all road records (read-only).
Step 11 logout.php Session destroyed. Redirect to ahome.php.

Authentication & Session Management

Both Citizens and Officers use PHP's native session mechanism

1. Login: $_SESSION['email'] = $user_data['email']; (set on successful login) Guard: Every protected page includes check_login($con) from the role's functions file.

2. If the session is not set, the user is immediately redirected to the login page.

3. Logout: unset($_SESSION['email']); header('Location: ahome.php');

Two separate check_login() functions exist

1. cfunctions.php - queries the 'citizen' table, redirects to clogin.php if unauthenticated.
   ofunctions.php - queries the 'officer' table, redirects to ologin.php if unauthenticated.

Setup & Deployment Guide

Requirements
MAMP (or XAMPP) with Apache + MySQL

1. PHP 8.x

2. Browser (Safari, Chrome, Firefox)

Installation Steps

Copy project folder to /Applications/MAMP/htdocs/Traffic-Monitoring-System/

Open MAMP and click 'Start Servers' (both Apache and MySQL must be green).

Open phpMyAdmin at http://localhost:8888/phpMyAdmin

Create a new database named 'project'.

Select 'project' database -> Import tab -> choose database.sql -> click Go.

Open the site at http://localhost:8888/Traffic-Monitoring-System/ahome.php
Database Credentials (aconnection.php)

Parameter Value

Host localhost

Username root
Password root (MAMP default)

Database project

Security Notes & Known Limitations

Issue Detail
SQL Injection Queries use raw string interpolation. Use prepared statements (PDO/MySQLi) in production.
Plain-text passwords Passwords stored as plain text. Use password_hash() and password_verify() in production.
No admin role There is no separate admin login. ahome.php is publicly accessible.
No input sanitization User input is not sanitised before DB insertion. Add htmlspecialchars/strip_tags.
Payment is simulated cpayment.php marks fine as paid but does not connect to a real payment gateway.
No CSRF protection Forms have no CSRF tokens. Add tokens to all POST forms in production.

Frontend Assets & Directory Layout

Directory Contents
assets/css/ Bootstrap, Font Awesome, Owl Carousel, Lightbox, main template CSS assets/js/ jQuery, Bootstrap JS, Owl Carousel, ScrollReveal, Lightbox, custom.js assets/images/ Slider images (a.jpg, b.jpg, c.jpg), icons, team photos
vendor/bootstrap/ Bootstrap 4 CSS + JS
vendor/animate/ Animate.css for scroll animations
vendor/select2/ Select2 dropdown library

vendor/perfect-scrollbar/ Smooth scrollbar for tables vendor/tilt/ Tilt.js - 3D tilt effect on login images
vendor/css-hamburgers/ Hamburger menu animation CSS css/ util.css + main.css - login page layout styles
font-awesome-4.7.0/ Icon font library
images/icons/ Favicon and SVG illustrations (citizen.svg, officer.svg, traffic.svg)

KEY DETAILS

1. Admin portal manages user accounts, officer assignments, road data, and signalling configurations.
2. Officer portal logs violators, records fines, and updates real-time road and signal status.
3. Citizen portal supports account registration, violation history lookup, and online fine payment.
4. Automated daily reports summarize traffic violations and flag roads needing infrastructure attention.
5. Built with PHP, MySQL (XAMPP), JavaScript, and CSS with a responsive multi-role front-end.